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Q1. Secant Method

$f(x) = x^3 - 5x - 9$

Initial given guess $x_0 = 2$, $x_1 = 3$

$f(x_0) = f(2) = (2)^3 - 5(2) - 9 = 8 - 10 - 9 = -11$

$f(x_1) = f(3) = (3)^3 - 5(3) - 9 = 27 - 15 - 9 = 3$

1st iteration:-

Secant Formula

$$x_{k+1} = \frac{x_{k-1} f(x_k) - x_k f(x_{k-1})}{f(x_k) - f(x_{k-1})}$$

Let $k = 1$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{(2)(3) - (3)(-11)}{(3) - (-11)}$$

$$= \frac{6 + 33}{3 + 11} = \frac{39}{14} = 2.7857143$$

$$\boxed{x_2 = 2.7857143}$$

$f(x_2) = (2.7857143)^3 - 5(2.7857143) - 9$

$= -1.3108598$

2nd iteration:-

Let $k = 2$

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{(3)(-1.3108598) - (2.7857143)(3)}{-1.3108598 - 3}$$

$\boxed{x_3 = 2.8508750}$, $f(x_3) = -0.0839219$

3rd iteration:-

Let $k = 3$

$$x_4 = \frac{x_2 f(x_3) - x_3 f(x_2)}{f(x_3) - f(x_2)}$$

$$= \frac{(2.7857143)(-0.0839219) - (2.8508750)(-1.3102598)}{(-0.0839219) - (-1.3108598}$$

$$\boxed{x_4 = 2.8553320}$$

$$f(x_4) = (2.8553320)^3 - 5(2.8553320) - 9 = 0.0026357$$

4th iteration:-

Let $k = 4$

$$x_5 = \frac{(x_3) f(x_4) - (x_4) f(x_3)}{f(x_4) - f(x_3)}$$

$$= \frac{(2.8508750)(0.0026357) - (2.8553320)(-0.0839219)}{0.0026357 - (-0.0839219)}$$

$$\boxed{x_5 = 2.8552260}$$

Q2. Jacobi Method

$10x + y + 2z = 12$

$2x + 10y + z = 13$

$2x + 2y + 10z = 14$

Initial guess

$x_0 = 0, y_0 = 0, z_0 = 0$

$|10| > |1| + |1| \Rightarrow 10 > 2$

$|10| > |1| + |2| \Rightarrow 10 > 3$

$|10| > |2| + |2| \Rightarrow 10 > 2$

So, our system of equation is diagonally dominated

$$x_{k+1} = \frac{12 - y_k - z_k}{10} \qquad y_{k+1} = \frac{13 - 2x_k - z_k}{10}$$

$$z_{k+1} = \frac{14 - 2x_k - 2y_k}{10}$$

First iteration:- $K = 0$

$$x_1 = \frac{12 - y_0 - z_0}{10}$$

$$x_1 = \frac{12 - 0 - 0}{10}$$

$$x_1 = 1.2$$

$$y_1 = \frac{13 - 2x_0 - z_0}{10}$$

$$y_1 = \frac{13 - 2(0) - 0}{10}$$

$$y_1 = 1.3$$

$$z_1 = \frac{14 - 2x_0 - 2y_0}{10}$$

$$z_1 = \frac{14 - 2(0) - 2(0)}{10}$$

$$z_1 = 1.4$$

$$\boxed{x_1 = 1.2,\ y_1 = 1.3,\ z_1 = 1.4}$$

2nd iteration:- K=1

$$x_2 = \frac{12 - y_1 - z_1}{10}$$

$$x_2 = \frac{12 - 1.3 - 1.4}{10}$$

$$x_2 = 0.93$$

$$y_2 = \frac{13 - 2x_1 - z_1}{10}$$

$$y_2 = \frac{13 - 2(1.2) - 1.4}{10}$$

$$y_2 = 0.921$$

$$z_2 = \frac{14 - 2x_1 - y_2}{10}$$

$$z_2 = \frac{14 - 2(1.2) - 1.32}{10}$$

$$z_2 = 0.90$$

$$x_2 = 0.93,\ y_2 = 0.92,\ z_2 = 0.90$$

3rd iteration:- K=2

$$x_3 = \frac{12 - y_2 - z_2}{10}$$

$$x_3 = \frac{12 - 0.92 - 0.9}{10}$$

$$x_3 = 1.018$$

$$y_3 = \frac{13 - 2(x_2) - z_2}{10}$$

$$y_3 = \frac{13 - 2(0.93) - 0.9}{10}$$

$$y_3 = 1.024$$

$$z_3 = \frac{14 - 2x_2 - 2y_2}{10}$$

$$z_3 = \frac{14 - 2(0.93) - 2(0.9}{10}$$

$$z_3 = 1.030$$

$$x_3 = 1.018,\ y_3 = 1.024,\ z_3 = 1.030$$

4th iteration:- K=3

$$x_4 = \frac{12 - y_3 - z_3}{10}$$

$$x_4 = \frac{12 - (1.024) - 1.030}{10}$$

$$x_4 = 0.9946$$

$$y_4 = \frac{13 - 2(x_3) - z_3}{10}$$

$$y_4 = \frac{13 - 2(1.018) - 1.03}{10}$$

$$y_4 = 0.9934$$

$$z_4 = \frac{14 - 2(x_3) - 2y_3}{10}$$

$$z_4 = \frac{14 - 2(1.018) - 2(1.0}{10}$$

$$z_4 = 0.9916$$

$$x_4 = 0.9946,\ y_4 = 0.9934,\ z_4 = 0.9910$$

Result after 4 iteration

$(x, y, z) = (0.9946, 0.9934, 0.9916)$

Q2. Gaus-Seidel Method

Equations:

$$x_{k+1} = \frac{12 - y_k - z_k}{10}, \quad y_{k+1} = \frac{13 - 2x_{k+1} - z_k}{10}$$

$$z_{k+1} = \frac{14 - 2x_{k+1} - 2y_{k+1}}{10}$$

1st iteration:- K=0

$$x_1 = \frac{12 - y_0 - z_0}{10}$$

$$x_1 = \frac{12 - 0 - 0}{10}$$

$$x_1 = 1.2$$

$$y_1 = \frac{13 - 2x_1 - z_0}{10}$$

$$y_1 = \frac{13 - 2(1.2) - 0}{10}$$

$$y_1 = 1.060$$

$$z_1 = \frac{14 - 2x_1 - 2y_1}{10}$$

$$z_1 = \frac{14 - 2(1.2) - 2(1.06)}{10}$$

$$z_1 = 0.948$$

$x_1 = 1.2, \; y_1 = 1.060, \; z_1 = 0.948$

2nd iteration:- K=1

$$x_2 = \frac{12 - y_1 - z_1}{10}$$

$$x_2 = \frac{12 - 1.060 - 0.948}{10}$$

$$x_2 = 0.9992$$

$$y_2 = \frac{13 - 2x_2 - z_1}{10}$$

$$y_2 = \frac{(13 - 2(0.9992) (-0.948)}{10}$$

$$y_2 = 1.00536$$

$$z_2 = \frac{14 - 2x_2 - 2y_2}{10}$$

$$z_2 = \frac{14 - 2(0.9992) - 2(1.00536)}{10}$$

$$z_2 = 0.999088$$

3rd iteration:- K=2

$$x_3 = \frac{12 - y_2 - z_2}{10} = \frac{12 - 1.00536 - 0.999088}{10} = 0.9995552$$

$$y_3 = \frac{13 - 2x_3 - z_2}{10} = \frac{13 - 2(0.9995552) - 0.9990880}{10}$$

$$y_3 = 1.0001802$$

$$z_3 = \frac{14 - 2x_3 - 2y_3}{10} = \frac{14 - 2(0.999552) - 2(1.0001802)}{10}$$

$$z_3 = 1.00000536$$

$x_3 = 0.9995552$, $y_3 = 1.0001802$, $z_3 = 1.00000536$

4th iteration:- K=3

$$x_4 = \frac{12 - y_3 - z_3}{10} = \frac{12 - (1.0001802) - (1.00000536)}{10}$$

$$x_4 = 0.9999767$$

$$y_4 = \frac{13 - 2x_4 - z_3}{10} = \frac{13 - 2(0.9999767) - 1.00000536}{10}$$

$$y_4 = 0.9999993$$

$$z_4 = \frac{14 - 2x_4 - 2y_4}{10} = \frac{14 - 2(0.9999767) - 2(0.9999993)}{10}$$

$$z_4 = 1.00000048$$

$x_4 = 0.9999767$, $y_4 = 0.9999993$, $z_4 = 1.00000048$

Q3

$10c_1 + c_2 + c_3 = 13$

$2c_1 + 10c_2 + 2c_3 = 14$

$2c_1 + 2c_2 + 10c_3 = 15$

Let

$|10| > |1| + |1| \Rightarrow 10 > 2$

$|10| > |2| + |2| \Rightarrow 10 > 4$

$|10| > |2| + |2| \Rightarrow 10 > 4$

So, our system is diagonally dominated

Initial guess:-

$(0, 0, 0)$

General Equations:-

$$c_1^{k+1} = \frac{13 - c_2^{(k)} - c_3^{(k)}}{10}, \quad c_2 = \frac{14 - 2c_3^{(k)} - 2c_1^{(k+1)}}{10}$$

$$c_3 = \frac{15 - 2c_1^{(k+1)} - 2c_2^{(k+1)}}{10}$$

first iteration (K=0)

$$C_1^{(1)} = \frac{13 - C_2^{(0)} - C_3^{(0)}}{10} = \frac{13-0-0}{10} = \frac{13}{10} = 1.3$$

$$C_2^{(1)} = \frac{14 - 2C_1^{(1)} - 2C_3^{(0)}}{10} = \frac{14 - 2(1.3) - 2(0)}{10} = \frac{14-2.6}{10} = 1.14$$

$$C_3^{(1)} = \frac{15 - 2C_2^{(1)} - 2C_1^{(1)}}{10} = \frac{15 - 2(1.14) - 2(1.3)}{10} = 1.012$$

$(C_1^{(1)} = 1.3,\ C_2^{(1)} = 1.14,\ C_3^{(1)} = 1.012)$

Second iteration (K=1)

$$C_1^{(2)} = \frac{13 - C_2^{(1)} - C_3^{(1)}}{10} = \frac{13 - 1.14 - 1.012}{10} = 1.0848$$

$$C_2^{(2)} = \frac{14 - 2C_1^{(2)} - 2C_3^{(1)}}{10} = \frac{14 - 2(1.0848) - 2(1.012)}{10} = \frac{14 - 2.1696 - 2.024}{10}$$

$$C_2^{(2)} = 0.98064$$

$$C_3^{(2)} = \frac{15 - 2C_2^{(2)} - 2C_3^{(2)}}{10} = \frac{15 - 2(1.048) - 2(1.012)}{10}$$

$$= 1.086912$$

$C_1^{(2)} = 1.0848,\ C_2^{(2)} = 0.98064,\ C_3^{(2)} = 1.086912$

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Third iteration (k=2)

$$C_1^{(3)} = \frac{13 - C_2^{(2)} - C_3^{(2)}}{10} = \frac{13 - 0.98064 - 1.086912}{10} = 1.0932448$$

$$C_2^{(3)} = \frac{14 - 2C_1^{(3)} - 2C_3^{(2)}}{10} = \frac{14 - 2(1.0932448) - 2(1.086912)}{10}$$

$$= 0.9639686$$

$$C_3^{(3)} = \frac{15 - 2C_1^{(3)} - 2C_2^{(3)}}{10} = \frac{15 - 2(1.0932448) - 2(0.9639686)}{10}$$

$$= 1.0885573$$

$$\left(C_1^{(3)} = 1.0932448,\ C_2^{(3)} = 0.9639686,\ C_3^{(3)} = 1.0885573\right)$$

Fourth iteration (k=3)

$$C_1^{(4)} = \frac{13 - C_2^{(3)} - C_3^{(3)}}{10} = \frac{13 - (0.9639686) - (1.0885573)}{10} = 1.0947474$$

$$C_2^{(4)} = \frac{14 - 2C_1^{(4)} - 2C_3^{(3)}}{10} = \frac{14 - 2(1.0947474) - 2(1.0885573)}{10}$$

$$= 0.9633391$$

$$C_3^{(4)} = \frac{15 - 2C_1^{(4)} - 2C_2^{(4)}}{10} = \frac{15 - 2(1.0947474) - 2(0.9633391)}{10}$$

$$= 1.0883827$$

$$\left(C_1^{(4)} = 1.0947474,\ C_2^{(4)} = 0.9633391,\ C_3^{(4)} = 1.0883827\right)$$

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